

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

SOFTWARE ENGINEERING

TWO MARKS QUESTION BANK

UNIT 1

1. State the IEEE definition of Software Engineering
2. Distinguish software product and process
3. What is software process?
4. Name any two software quality attributes and define it
5. Name the basic techniques used by the software engineering techniques to handle complexity in a problem
6. Draw a schematic diagram to represent the iterative waterfall model of software development. On the diagram, represent the deliverables that need to be produced at the end of each phase.
7. What are the objectives of the feasibility study phase of software development?
8. Define the term – Phase containment of errors
9. Give the major advantages of Object-Oriented Design (OOD) methodologies
10. Define computer system engineering
11. What is a prototype? Give its advantages
12. What is a Phase entry and exit criterion?
13. Differentiate Control-flow based design and Data-flow based design

UNIT II

1. Write short notes on risk components
2. Create a task set for project planning
3. What are the steps in risk management?
4. What is work product?
5. List the major responsibilities of a software project manager
6. What is meant by the size of a software project? Give its importance
7. What is the purpose of Activity network and the Gantt Chart
8. Define software configuration management
9. Write about Work Breakdown structure
10. Define algebraic and axiomatic specification methods
11. Differentiate Staffing and Scheduling
12. Compare LOC and FP
13. Brief the Configuration Management Activities
14. What would be the organization of the SRS document?

UNIT III

1. List the characteristics of a good software design
2. Compare object-oriented and function oriented system
3. Define coupling & Cohesion
4. How can functional independence in a software design be achieved?

5. What are the advantages of layered software design?
6. What is meant by balancing a DFD?
7. What are the main shortcomings of DFD as tool for performing structuring analysis?
8. Differentiate between structured analysis and structured design in the context of function-oriented design
9. Differentiate between structure chart and a flow chart as design representation techniques
10. How a DFD model can be transformed into Structure Chart?
11. Brief about a layered software design
12. Differentiate DFD and Structure Chart
13. Define Functional Independence

UNIT IV

1. Define UML
2. List the advantages of object-oriented design
3. Brief the basic elements of the state chart diagram with example
4. Difference between sequence and collaboration diagram.
5. Draw the use case diagram for a online banking system
6. Differentiate Patterns and Anti-Patterns
7. Compare the three different types of Patterns
8. Differentiate Entity Objects and Controller objects
9. List the OOD Goodness criteria
10. Brief about unified process

UNIT V

1. Define component and component based system
2. What is unit testing?
3. List the characteristics of a good user interface.
4. Comment on alpha and beta testing.
5. What are the types of user interfaces ? Explain them shortly in a line
6. Compare Equivalence Class Partitioning and Boundary value analysis
7. Differentiate Fault-based Testing and Coverage-based testing
8. Brief about the program analysis tools
9. Differentiate System testing and Integration Testing
10. Define Gunning Fog Index

TEN MARKS QUESTION BANK

UNIT 1

1. Describe Water fall model with neat sketch [April 2016] [Nov 2015]
2. Explain the concepts involved in Spiral Model [April 2016]
3. Compare all software development lifecycle models
4. Explain the emergence of software engineering

UNIT II

1. Discuss the Software Configuration Management process[April 2016]
2. Elaborate requirements elicitation with an example [April 2016]
3. a) What are the responsibilities of a software project manager? Explain (7) [Nov 2015]
b) Write short notes on COCOMO Model (4)
4. Explain in detail about software project scheduling [Nov 2015]
5. a) Differentiate Functional and Non-functional requirements.
b) Brief about the techniques for representing complex logic.
6. Elaborate Staffing techniques.
7. a) What are the different types of software projects?
b) Write short notes on Empirical estimation techniques
8. Explain in detail about Risk Management.

UNIT III

1. Tamil Nadu Electricity Board (TNEB) would like to automate its billing process. Customers apply for a connection (domestic/commercial). EB Staff take readings and update the system. Each customer is required to pay charges bi-monthly according to the rates for the type of connection. Customers can choose to pay either by cash/card. A bill is generated on payment. Monthly reports are provided to the EB Manager. [April 2016]
 - a. Draw the Level-0 DFD
 - b. Draw the Level-1 DFD
2. Describe the concepts of Coupling & Cohesion and its types [April 2016].
3. Explain how to develop the DFD model of a system with an example
4. Explain the various approaches to software design with examples

UNIT IV

1. Demonstrate the concept of Use Case Diagram with suitable example[April 2016]
2. What is Domain Modeling? Explain
3. What is Pattern? What are the pros and cons of design patterns? Explain any two of the patterns?

UNIT V

1. Illustrate User Interface Design steps [Nov 2015]
2. Discuss black box testing with neat sketch [April 2016] [Nov 2015]
3. A) What are the characteristics of a good user interface? Explain. (7).
B) State some rules for software documentation (4) [April 2016]